MKSSS's
Cummins College of Engineering for Women, Pune
(An Autonomous Institute Affiliated to SPPU)

Course: Design and Analysis of Algorithms (23PCIT501L)
Academic Year: 2026-27

DESIGN AND ANALYSIS OF ALGORITHMS PARALLEL PROGRAMMING PROJECT

OUTPUT DOCUMENT

Problem Statement: Portfolio Optimization using Branch and Bound and OpenMP

submitted by
Group no: 01 (B4-Batch)

UID	Student Name
UIT2024966	TRUPTI METE
UIT2024967	TEJAL UDGAVE
UIT2024968	VAISHNAVI KARVE
UIT2024261	KALYANI SOMVANSHI

Under the guidance of:
Prof. Harshad Suryakant Wadkar
Assistant Professor (I. T.)
Cummins College of Engineering for Women, Pune

1. EXECUTION OVERVIEW

This document presents the program output (terminal screenshots) of the four programs developed for the project: the sequential exhaustive search, the OpenMP exhaustive search, the sequential Branch and Bound and the OpenMP Branch and Bound. All runs use 20 stocks, an allocation unit of 5%, a maximum of 20% per stock and a daily risk limit of 0.016. Table 1 lists the runs and the figure showing each one.

Program	Threads	Execution time (s)	Figure	Section
Sequential exhaustive search	Sequential	5577.673371	1	2
OpenMP exhaustive search	1	1129.855748	2	3
OpenMP exhaustive search	4	723.258840	3	3
OpenMP exhaustive search	6	540.432622	4	3
OpenMP exhaustive search	8	508.497909	5	3
OpenMP exhaustive search	10	490.809710	6	3
OpenMP exhaustive search	16	395.482028	7	3
Sequential Branch and Bound	Sequential	12.8467	8, 9	4
OpenMP Branch and Bound	1	7.13065	10	5
OpenMP Branch and Bound	4	6.82983	11	5
OpenMP Branch and Bound	8	4.27121	12	5
OpenMP Branch and Bound	10	3.56736	13	5
OpenMP Branch and Bound	16	2.60595	14	5

Table 1: *Runs shown in the output screenshots*

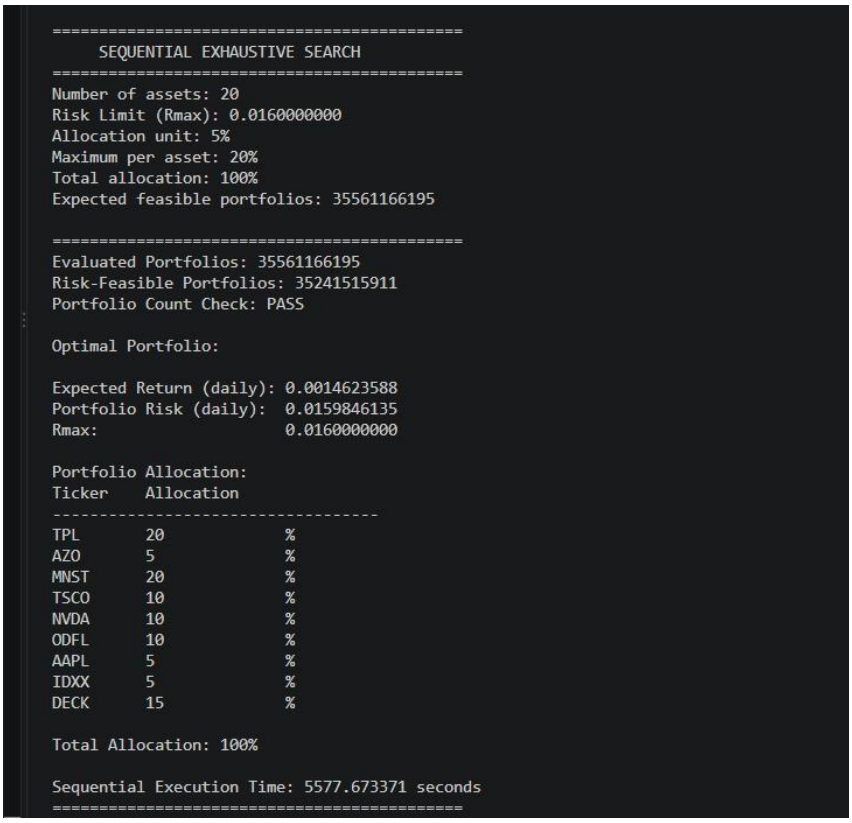
Every run reports the same optimal portfolio (Table 2): expected daily return 0.00146236, daily risk 0.01598461 (limit 0.016) and a total allocation of 100%.

Stock	Weight	Stock	Weight	Stock	Weight
TPL	20%	NVDA	10%	DECK	15%
MNST	20%	ODFL	10%	AZO	5%
TSCO	10%	AAPL	5%	IDXX	5%

Table 2: *Optimal portfolio reported by all runs (all other stocks 0%)*

2. SEQUENTIAL EXHAUSTIVE SEARCH

The sequential exhaustive search evaluates all 35,561,166,195 feasible portfolios and passes the portfolio-count check.



```
========================================
     SEQUENTIAL EXHAUSTIVE SEARCH
========================================
Number of assets: 20
Risk Limit (Rmax): 0.0160000000
Allocation unit: 5%
Maximum per asset: 20%
Total allocation: 100%
Expected feasible portfolios: 35561166195

========================================
Evaluated Portfolios: 35561166195
Risk-Feasible Portfolios: 35241515911
Portfolio Count Check: PASS

Optimal Portfolio:

Expected Return (daily): 0.0014623588
Portfolio Risk (daily):  0.0159846135
Rmax:                    0.0160000000

Portfolio Allocation:
Ticker    Allocation
----------------------------------
TPL       20             %
AZO       5              %
MNST      20             %
TSCO      10             %
NVDA      10             %
ODFL      10             %
AAPL      5              %
IDXX      5              %
DECK      15             %

Total Allocation: 100%

Sequential Execution Time: 5577.673371 seconds
========================================
```

Figure 1: *Sequential exhaustive search output*

3. OPENMP EXHAUSTIVE SEARCH

The OpenMP exhaustive search takes the number of stocks and the number of threads as command-line arguments (./bf_openmp 20 <threads>). Each run evaluates all 35,561,166,195 feasible portfolios (Exhaustive check: PASS).

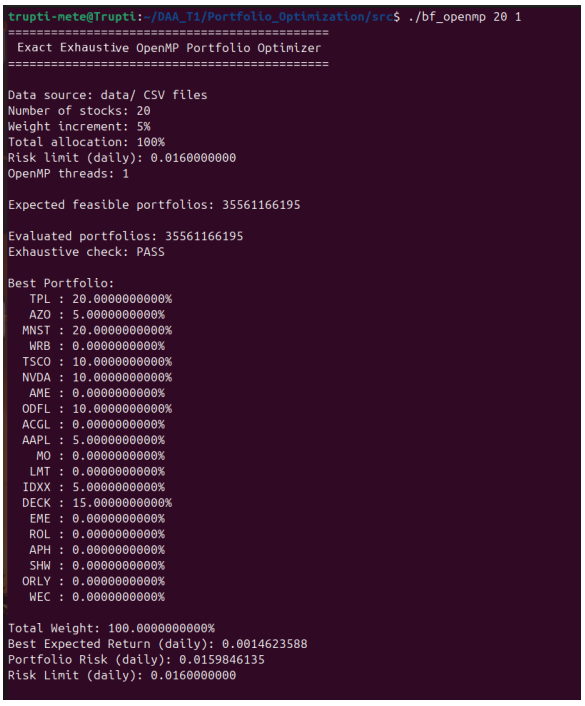


Figure 2: *OpenMP exhaustive search, 1 thread*

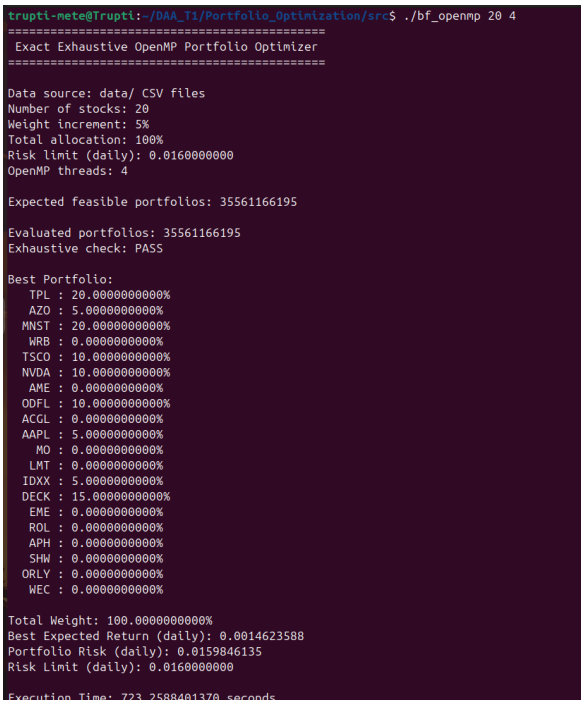


Figure 3: *OpenMP exhaustive search, 4 threads*

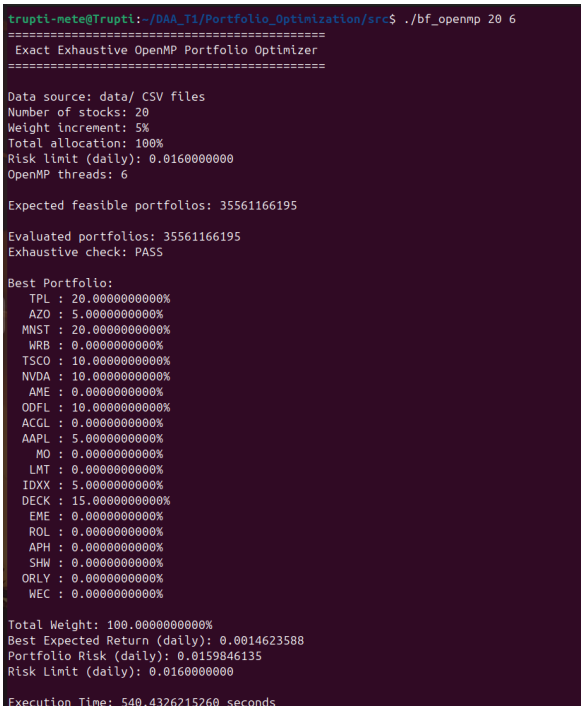


Figure 4: *OpenMP exhaustive search, 6 threads*

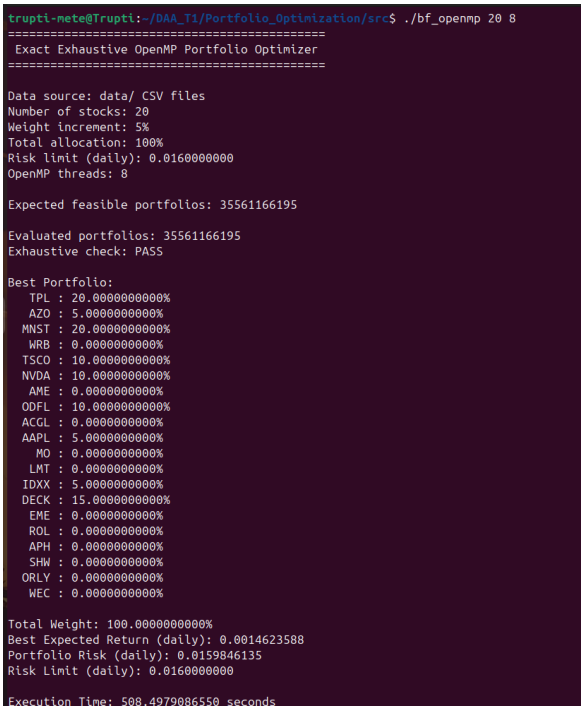


Figure 5: *OpenMP exhaustive search, 8 threads*

```
trupti-mete@Trupti:~/DAA_T1/Portfolio_Optimization/src$ ./bf_openmp 20 10
============================================
 Exact Exhaustive OpenMP Portfolio Optimizer
============================================

Data source: data/ CSV files
Number of stocks: 20
Weight increment: 5%
Total allocation: 100%
Risk limit (daily): 0.0160000000
OpenMP threads: 10

Expected feasible portfolios: 35561166195

Evaluated portfolios: 35561166195
Exhaustive check: PASS

Best Portfolio:
   TPL : 20.0000000000%
   AZO : 5.0000000000%
  MNST : 20.0000000000%
   WRB : 0.0000000000%
  TSCO : 10.0000000000%
  NVDA : 10.0000000000%
   AME : 0.0000000000%
  ODFL : 10.0000000000%
  ACGL : 0.0000000000%
  AAPL : 5.0000000000%
    MO : 0.0000000000%
   LMT : 0.0000000000%
  IDXX : 5.0000000000%
  DECK : 15.0000000000%
   EME : 0.0000000000%
   ROL : 0.0000000000%
   APH : 0.0000000000%
   SHW : 0.0000000000%
  ORLY : 0.0000000000%
   WEC : 0.0000000000%

Total Weight: 100.0000000000%
Best Expected Return (daily): 0.0014623588
Portfolio Risk (daily): 0.0159846135
Risk Limit (daily): 0.0160000000

Execution Time: 490.8097103590 seconds
```

Figure 6: *OpenMP exhaustive search, 10 threads*

```
trupti-mete@Trupti:~/DAA_T1/Portfolio_Optimization/src$ ./bf_openmp 20 16
============================================
 Exact Exhaustive OpenMP Portfolio Optimizer
============================================

Data source: data/ CSV files
Number of stocks: 20
Weight increment: 5%
Total allocation: 100%
Risk limit (daily): 0.0160000000
OpenMP threads: 16

Expected feasible portfolios: 35561166195

Evaluated portfolios: 35561166195
Exhaustive check: PASS

Best Portfolio:
   TPL : 20.0000000000%
   AZO : 5.0000000000%
  MNST : 20.0000000000%
   WRB : 0.0000000000%
  TSCO : 10.0000000000%
  NVDA : 10.0000000000%
   AME : 0.0000000000%
  ODFL : 10.0000000000%
  ACGL : 0.0000000000%
  AAPL : 5.0000000000%
    MO : 0.0000000000%
   LMT : 0.0000000000%
  IDXX : 5.0000000000%
  DECK : 15.0000000000%
   EME : 0.0000000000%
   ROL : 0.0000000000%
   APH : 0.0000000000%
   SHW : 0.0000000000%
  ORLY : 0.0000000000%
   WEC : 0.0000000000%

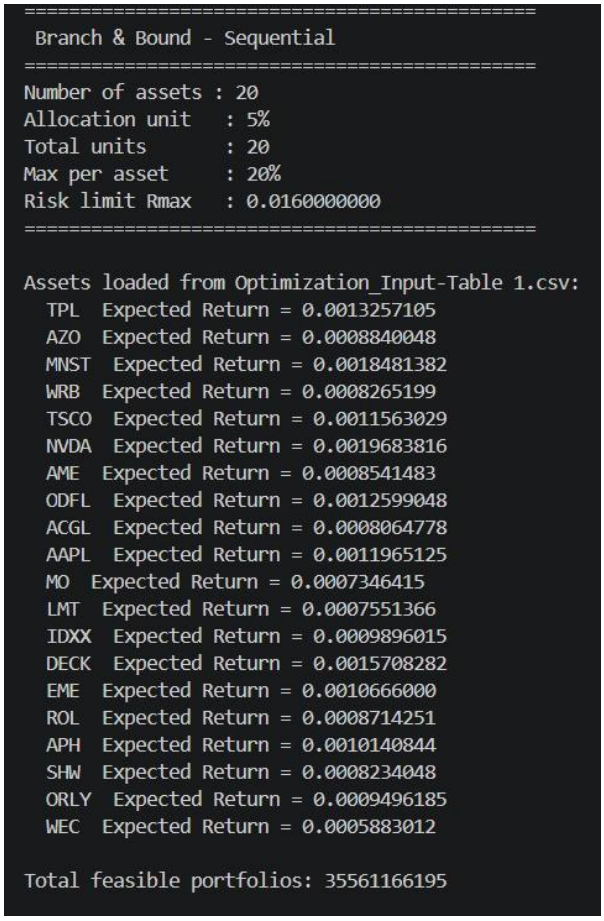
Total Weight: 100.0000000000%
Best Expected Return (daily): 0.0014623588
Portfolio Risk (daily): 0.0159846135
Risk Limit (daily): 0.0160000000

Execution Time: 395.4820279900 seconds
```

Figure 7: *OpenMP exhaustive search, 16 threads*

4. SEQUENTIAL BRANCH AND BOUND

The sequential Branch and Bound loads the expected returns from the input file, searches the tree with pruning and prints the search statistics: 24,265,951 nodes visited, 5,015,033 complete portfolios tested, 35,556,151,162 candidate portfolios pruned and a search-space reduction of 99.9859%. The candidate accounting check passes.

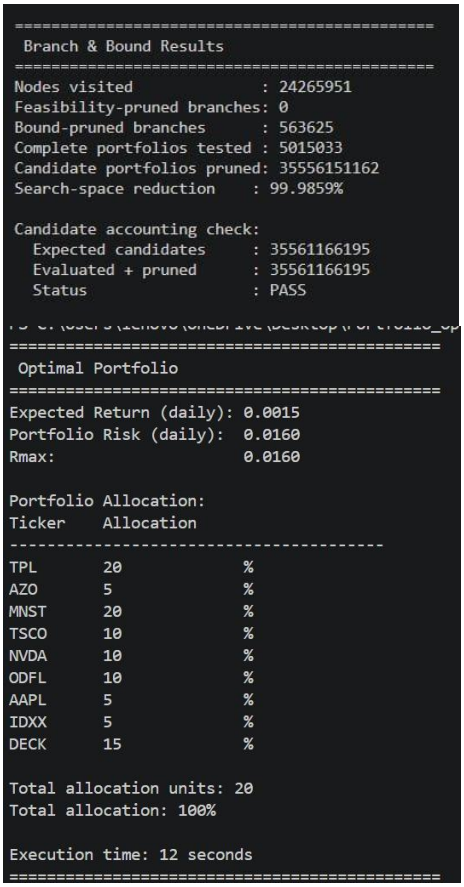


```
==========================================
 Branch & Bound - Sequential
==========================================
Number of assets : 20
Allocation unit   : 5%
Total units       : 20
Max per asset     : 20%
Risk limit Rmax   : 0.0160000000
==========================================

Assets loaded from Optimization_Input-Table 1.csv:
  TPL  Expected Return = 0.0013257105
  AZO  Expected Return = 0.0008840048
  MNST  Expected Return = 0.0018481382
  WRB  Expected Return = 0.0008265199
  TSCO  Expected Return = 0.0011563029
  NVDA  Expected Return = 0.0019683816
  AME  Expected Return = 0.0008541483
  ODFL  Expected Return = 0.0012599048
  ACGL  Expected Return = 0.0008064778
  AAPL  Expected Return = 0.0011965125
  MO  Expected Return = 0.0007346415
  LMT  Expected Return = 0.0007551366
  IDXX  Expected Return = 0.0009896015
  DECK  Expected Return = 0.0015708282
  EME  Expected Return = 0.0010666000
  ROL  Expected Return = 0.0008714251
  APH  Expected Return = 0.0010140844
  SHW  Expected Return = 0.0008234048
  ORLY  Expected Return = 0.0009496185
  WEC  Expected Return = 0.0005883012

Total feasible portfolios: 35561166195
```

Figure 8: *Sequential Branch and Bound: configuration and input data*



```
==========================================
 Branch & Bound Results
==========================================
Nodes visited              : 24265951
Feasibility-pruned branches: 0
Bound-pruned branches      : 563625
Complete portfolios tested : 5015033
Candidate portfolios pruned: 35556151162
Search-space reduction     : 99.9859%

Candidate accounting check:
  Expected candidates      : 35561166195
  Evaluated + pruned       : 35561166195
  Status                   : PASS

==========================================
 Optimal Portfolio
==========================================
Expected Return (daily): 0.0015
Portfolio Risk (daily):  0.0160
Rmax:                    0.0160

Portfolio Allocation:
Ticker    Allocation
----------------------------------------
TPL       20             %
AZO       5              %
MNST      20             %
TSCO      10             %
NVDA      10             %
ODFL      10             %
AAPL      5              %
IDXX      5              %
DECK      15             %

Total allocation units: 20
Total allocation: 100%

Execution time: 12 seconds
==========================================
```

Figure 9: *Sequential Branch and Bound: search statistics, optimal portfolio and execution time*

5. OPENMP BRANCH AND BOUND

The OpenMP Branch and Bound takes the number of stocks as an argument and the number of threads from OMP_NUM_THREADS (the 10-thread run uses the default of the machine). Every run reports Risk Constraint: PASS, Total Allocation: 100% and Allocation Constraint: PASS.

```
kalyanisomvanshi@Kalyanis-MacBook-Air Portfolio_Optimization % OMP_NUM_THREADS=1 ./branch_bound_parallel 20
Problem Size: 20 stocks
OpenMP Threads: 1

Best Expected Return: 0.00146236
Risk: 0.01598461

Portfolio:
  NVDA : 10.00000000%
  MNST : 20.00000000%
  DECK : 15.00000000%
   TPL : 20.00000000%
  ODFL : 10.00000000%
  AAPL : 5.00000000%
  TSCO : 10.00000000%
   EME : 0.00000000%
   APH : 0.00000000%
  IDXX : 5.00000000%
  ORLY : 0.00000000%
   AZO : 5.00000000%
   ROL : 0.00000000%
   AME : 0.00000000%
   WRB : 0.00000000%
   SHW : 0.00000000%
  ACGL : 0.00000000%
   LMT : 0.00000000%
    MO : 0.00000000%
   WEC : 0.00000000%

Execution Time: 7.13065000 seconds
Verified Risk: 0.01598461
Risk Constraint: PASS
Total Allocation: 100%
```

Figure 10: *OpenMP Branch and Bound, 1 thread (OMP_NUM_THREADS=1)*

```
kalyanisomvanshi@Kalyanis-MacBook-Air Portfolio_Optimization % OMP_NUM_THREADS=4 ./branch_bound_parallel 20
Problem Size: 20 stocks
OpenMP Threads: 4

Best Expected Return: 0.00146236
Risk: 0.01598461

Portfolio:
  NVDA : 10.00000000%
  MNST : 20.00000000%
  DECK : 15.00000000%
   TPL : 20.00000000%
  ODFL : 10.00000000%
  AAPL : 5.00000000%
  TSCO : 10.00000000%
   EME : 0.00000000%
   APH : 0.00000000%
  IDXX : 5.00000000%
  ORLY : 0.00000000%
   AZO : 5.00000000%
   ROL : 0.00000000%
   AME : 0.00000000%
   WRB : 0.00000000%
   SHW : 0.00000000%
  ACGL : 0.00000000%
   LMT : 0.00000000%
    MO : 0.00000000%
   WEC : 0.00000000%

Execution Time: 6.82982500 seconds
Verified Risk: 0.01598461
Risk Constraint: PASS
Total Allocation: 100%
Allocation Constraint: PASS
kalyanisomvanshi@Kalyanis-MacBook-Air Portfolio_Optimization %
```

Figure 11: *OpenMP Branch and Bound, 4 threads (OMP_NUM_THREADS=4)*

```
kalyanisomvanshi@Kalyanis-MacBook-Air Portfolio_Optimization % OMP_NUM_THREADS=8 ./branch_bound_parallel 20
Problem Size: 20 stocks
OpenMP Threads: 8

Best Expected Return: 0.00146236
Risk: 0.01598461

Portfolio:
  NVDA : 10.00000000%
  MNST : 20.00000000%
  DECK : 15.00000000%
   TPL : 20.00000000%
  ODFL : 10.00000000%
  AAPL : 5.00000000%
  TSCO : 10.00000000%
   EME : 0.00000000%
   APH : 0.00000000%
  IDXX : 5.00000000%
  ORLY : 0.00000000%
   AZO : 5.00000000%
   ROL : 0.00000000%
   AME : 0.00000000%
   WRB : 0.00000000%
   SHW : 0.00000000%
  ACGL : 0.00000000%
   LMT : 0.00000000%
    MO : 0.00000000%
   WEC : 0.00000000%

Execution Time: 4.27121000 seconds
Verified Risk: 0.01598461
Risk Constraint: PASS
Total Allocation: 100%
Allocation Constraint: PASS
kalyanisomvanshi@Kalyanis-MacBook-Air Portfolio_Optimization %
```

Figure 12: *OpenMP Branch and Bound, 8 threads (OMP_NUM_THREADS=8)*



Figure 13: *OpenMP Branch and Bound, 10 threads (default)*

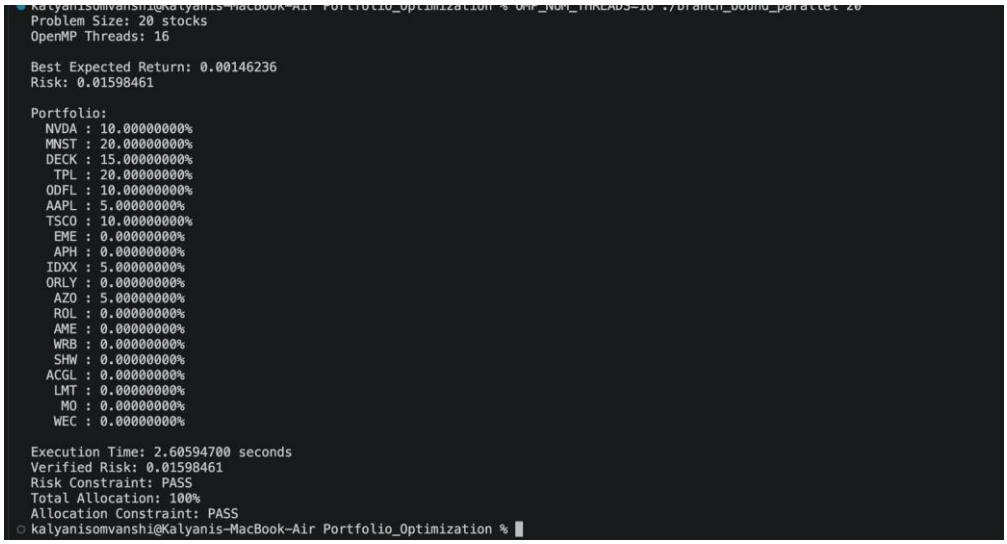


Figure 14: *OpenMP Branch and Bound, 16 threads (OMP_NUM_THREADS=16)*